Fibonacci Number

Question

The Fibonacci numbers, commonly denoted F(n) form a sequence, called the Fibonacci sequence, such that each number is the sum of the two preceding ones, starting from 0 and 1.

That is,

F(0) = 0, F(1) = 1

F(n) = F(n - 1) + F(n - 2), for n > 1.

Given n, calculate F(n).

Constraints

0 <= n <= 30

Solution

for doing this question you have to know about how recursion works properly... here i do it by recursion method.

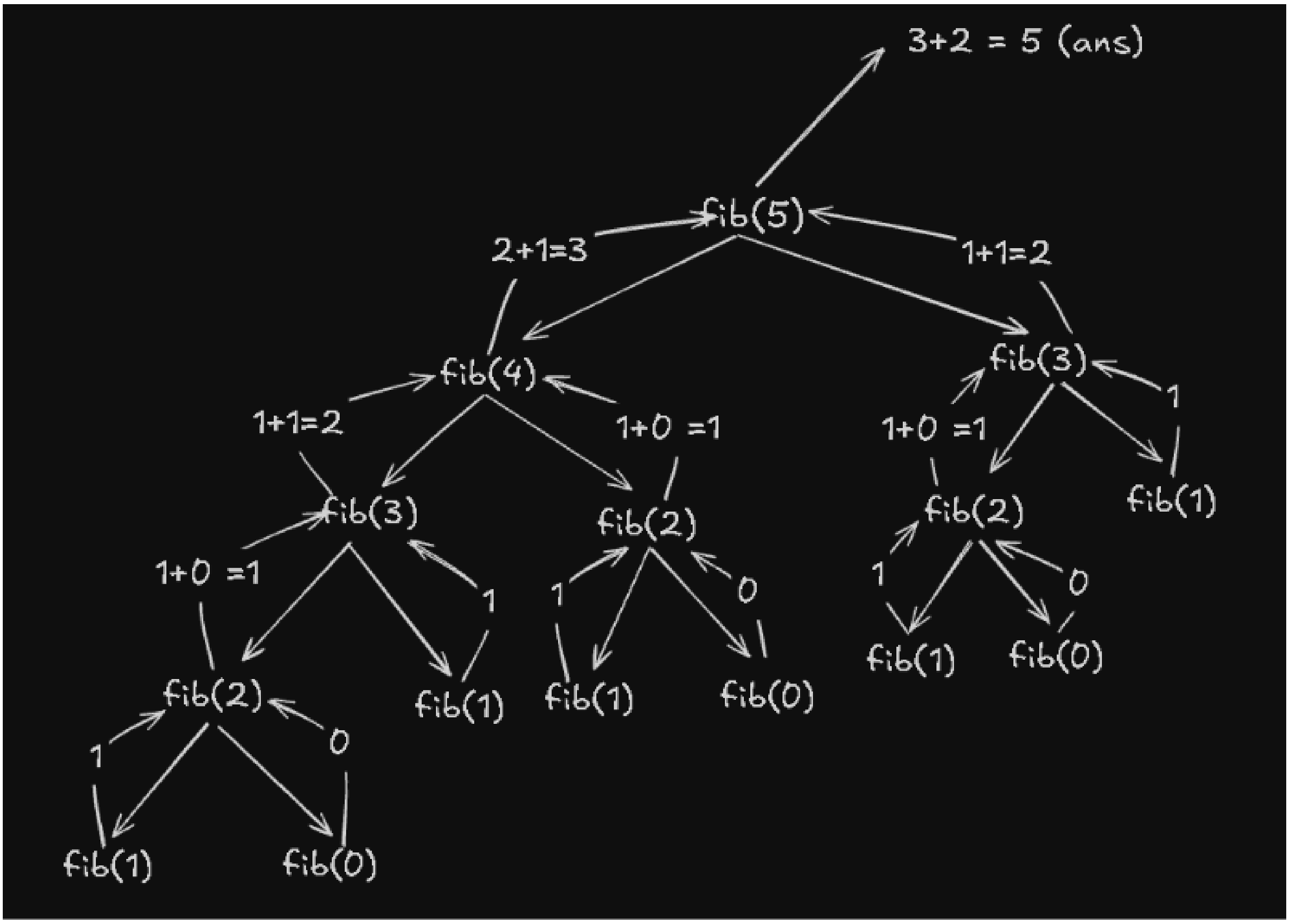
Now for recursion we need a base condition and here base condition is given as --

- F(0) = 0
- F(1) = 1

Pseudo code

```
fib(n){
  if(n==0 or n==1){
    return n
  }
  else{
    return fib(n-1)+fib(n-2)
  }
}
```

Recursion Tree (n = 5)



Code

```
class Solution {
    public int fib(int n) {
        if(n==0 || n==1){   // java or operator --> ||
            return n;
        }else{
            return fib(n-1)+fib(n-2);
        }


    }
}
```